

HW Class 6 (R Functions)

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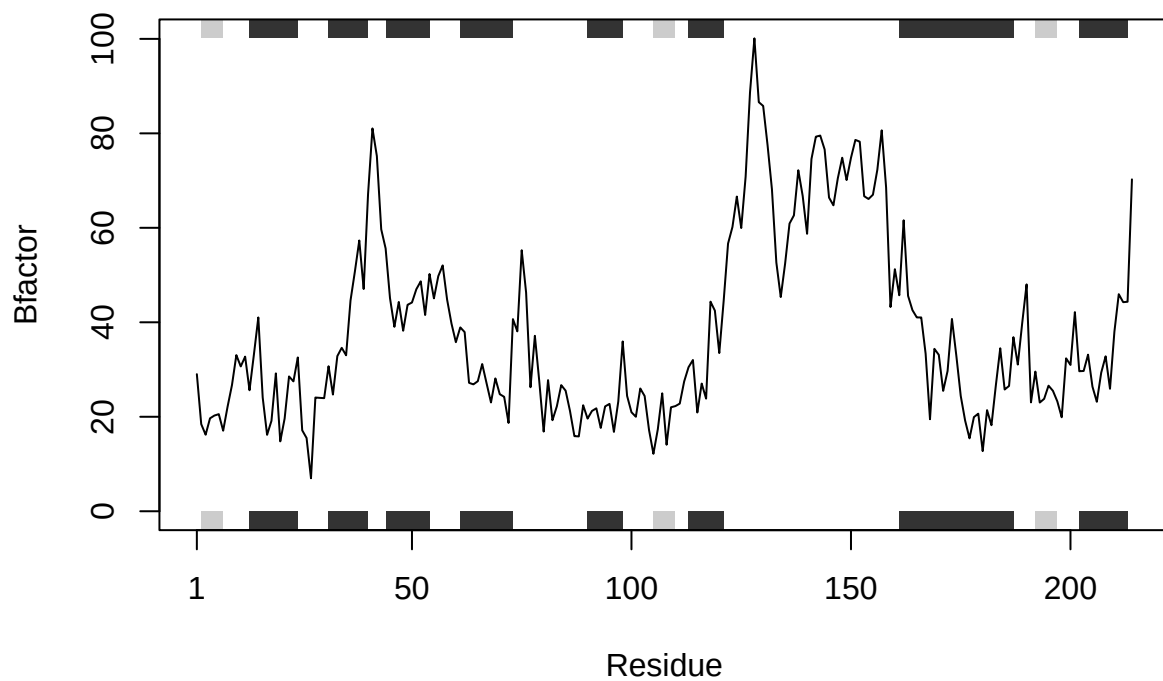
Original code

```
library(bio3d)
s1 <- read.pdb("4AKE") # kinase with drug

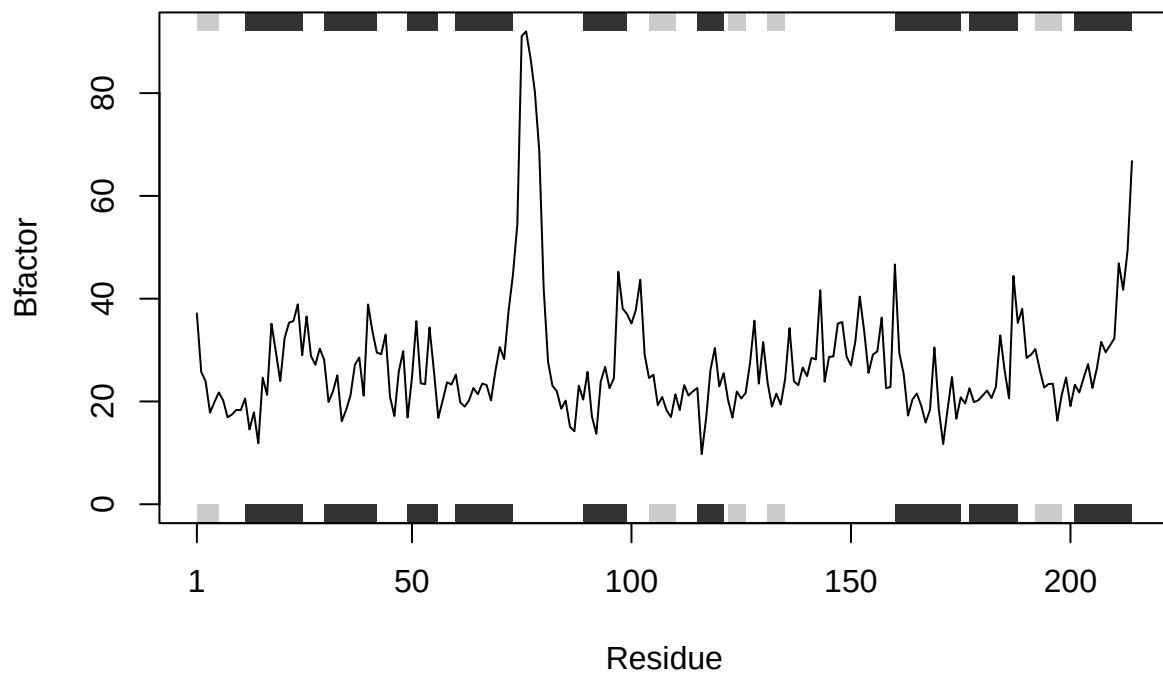
## Note: Accessing on-line PDB file
s2 <- read.pdb("1AKE") # kinase no drug

## Note: Accessing on-line PDB file
## PDB has ALT records, taking A only, rm.alt=TRUE
s3 <- read.pdb("1E4Y") # kinase with drug

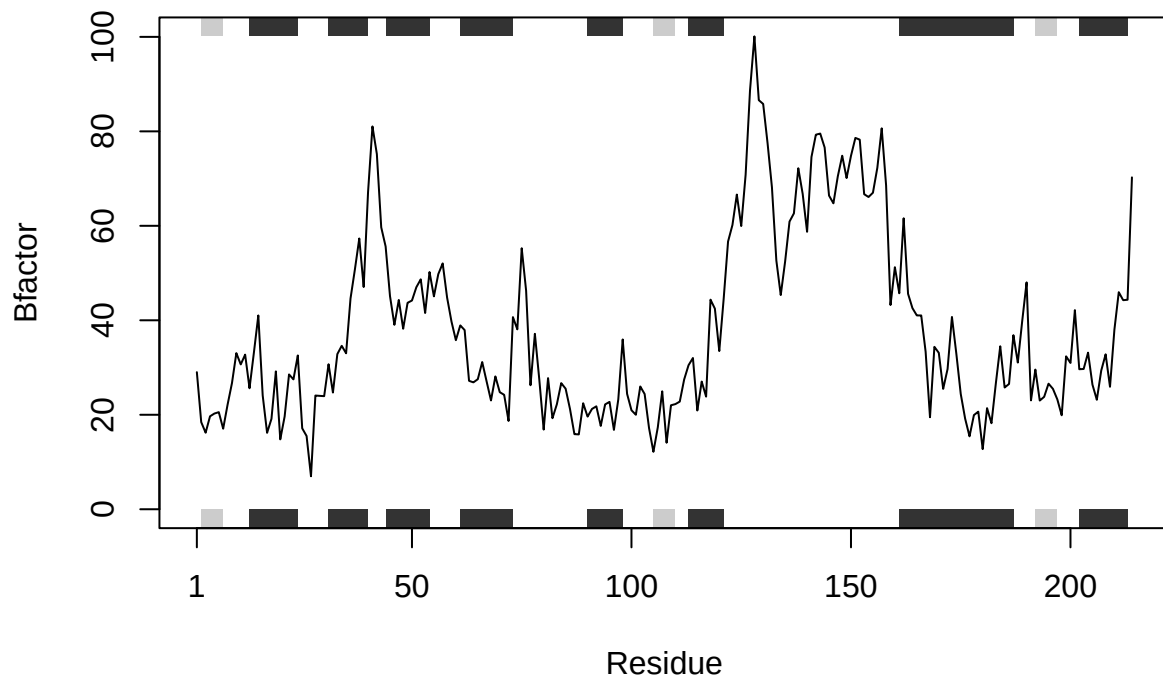
## Note: Accessing on-line PDB file
s1.chainA <- trim.pdb(s1, chain="A", elety="CA")
s2.chainA <- trim.pdb(s2, chain="A", elety="CA")
# there is a small error in original code
#where s1 is used instead of s3 below
s3.chainA <- trim.pdb(s1, chain="A", elety="CA")
s1.b <- s1.chainA$atom$b
s2.b <- s2.chainA$atom$b
s3.b <- s3.chainA$atom$b
plotb3(s1.b, sse=s1.chainA, typ="l", ylab="Bfactor")
```



```
plotb3(s2.b, sse=s2.chainA, typ="l", ylab="Bfactor")
```



```
plotb3(s3.b, sse=s3.chainA, typ="l", ylab="Bfactor")
```



Homework Class 6

Q6: How would you generalize the original code above to work with any set of input protein structures?

```
library(bio3d)

# the plot_protein() function reads a protein structure from the PDB database
# and extracts B-factor values and generates a plot for visualization of these
# values
# the inputs include: pdb_id, chain, and atom_type.
# pdb_id is a character that specifies the PDB ID, the chain is the protein chain
# additionally, the atom_type is the atom type to extract
# the outputs include a plot of B-factors and residue position as well as
# a numeric vector of the B-factor values

plot_protein <- function(pdb_id, chain = "A", atom_type = "CA") {

  # the PDB file is read
  pdb <- read.pdb(pdb_id)

  # the trim function is used to trim to chain A and alpha carbons
  pdb_trim <- trim.pdb(pdb, chain = chain, elety = atom_type)

  # the b-factor values is extracted from the trimmed structure
  b_factors <- pdb_trim$atom$b
```

```

# b-factors are plotted across residues
# the typ = "l" creates a line plot
# the sse adds secondary structure such as helix and sheet regions
plotb3(b_factors, sse = pdb_trim, typ = "l", ylab = "Bfactor")

# return values of the b-factor values
return(b_factors)
}

```

Example output run

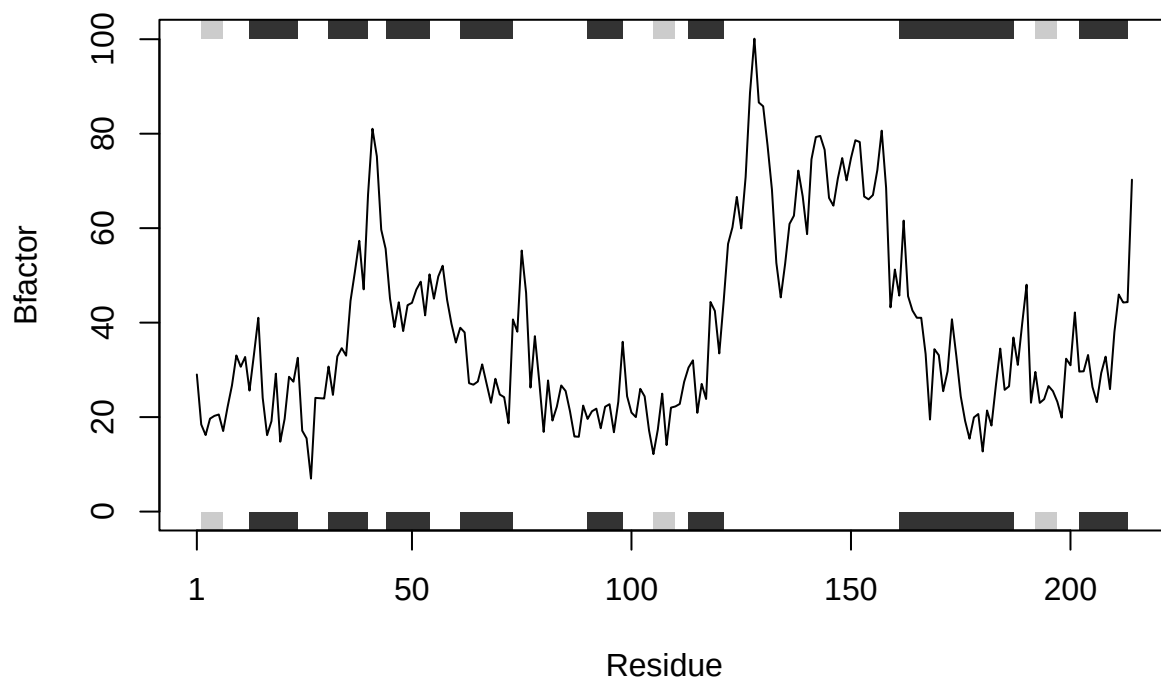
```
plot_protein("4AKE")
```

```
## Note: Accessing on-line PDB file
```

```
## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
```

```
## /var/folders/1x/9jyvq1ns09n6_zjnmq3xgx5h0000gn/T//RtmpzmuEmj/4AKE.pdb exists.
```

```
## Skipping download
```



```

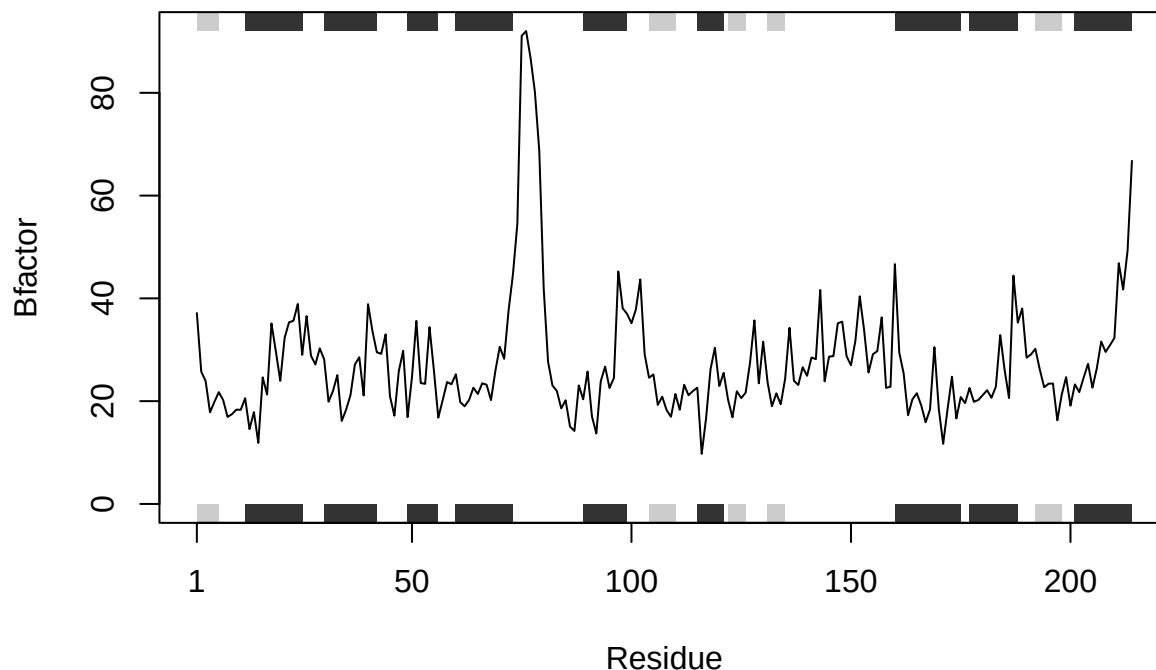
## [1] 29.02 18.44 16.20 19.67 20.26 20.55 17.05 22.13 26.71 33.05
## [11] 30.66 32.73 25.61 33.19 41.03 24.09 16.18 19.14 29.19 14.79
## [21] 19.63 28.54 27.49 32.56 17.13 15.50 6.98 24.07 24.00 23.94
## [31] 30.70 24.70 32.84 34.60 33.01 44.60 50.74 57.32 47.04 67.13
## [41] 81.04 75.20 59.68 55.63 45.12 39.04 44.31 38.21 43.70 44.19
## [51] 47.00 48.67 41.54 50.22 45.07 49.77 52.04 44.82 39.75 35.79
## [61] 38.92 37.93 27.18 26.86 27.53 31.16 27.08 23.03 28.12 24.78

```

```
## [71] 24.22 18.69 40.67 38.08 55.26 46.29 26.25 37.14 27.50 16.86
## [81] 27.76 19.27 22.22 26.70 25.52 21.22 15.90 15.84 22.44 19.61
## [91] 21.23 21.79 17.64 22.19 22.73 16.80 23.25 35.95 24.42 20.96
## [101] 20.00 25.99 24.39 17.19 12.16 17.35 24.97 14.08 22.01 22.26
## [111] 22.78 27.47 30.49 32.02 20.90 27.03 23.84 44.37 42.47 33.48
## [121] 44.56 56.67 60.18 66.62 59.95 70.81 88.63 100.11 86.60 85.80
## [131] 77.48 68.13 52.66 45.34 52.43 60.90 62.64 72.19 66.75 58.73
## [141] 74.57 79.29 79.53 76.58 66.40 64.76 70.48 74.84 70.11 74.82
## [151] 78.61 78.24 66.70 66.10 67.01 72.28 80.64 68.54 43.23 51.24
## [161] 45.72 61.60 45.61 42.57 41.03 41.02 33.34 19.48 34.38 33.11
## [171] 25.48 29.68 40.71 32.91 24.41 19.20 15.43 19.93 20.66 12.72
## [181] 21.40 18.21 26.68 34.50 25.77 26.52 36.85 31.05 39.84 48.03
## [191] 23.04 29.57 23.00 23.80 26.59 25.49 23.25 19.89 32.37 30.97
## [201] 42.16 29.64 29.69 33.15 26.38 23.17 29.35 32.80 25.92 38.01
## [211] 45.95 44.26 44.35 70.26
```

```
plot_protein("1AKE")
```

```
## Note: Accessing on-line PDB file
## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
## /var/folders/1x/9jyvqlns09n6_zjnmq3xgx5h0000gn/T//RtmpzmuEmj/1AKE.pdb exists.
## Skipping download
## PDB has ALT records, taking A only, rm.alt=TRUE
```



```
## [1] 37.14 25.76 23.90 17.83 19.86 21.75 20.21 16.92 17.47 18.35 18.31 20.57
## [13] 14.56 17.87 11.87 24.63 21.29 35.13 29.68 23.96 32.34 35.34 35.64 38.91
```

```
## [25] 29.00 36.55 28.83 27.15 30.28 28.13 19.90 21.95 25.07 16.15 18.35 21.19
## [37] 27.13 28.55 21.10 38.88 33.63 29.51 29.21 33.01 20.92 17.17 25.84 29.80
## [49] 16.89 24.66 35.62 23.52 23.37 34.41 25.96 16.79 20.20 23.72 23.29 25.23
## [61] 19.81 19.00 20.21 22.62 21.40 23.47 23.20 20.21 25.90 30.58 28.25 37.60
## [73] 44.66 54.46 91.10 92.02 86.85 80.21 68.72 42.01 27.69 23.06 21.98 18.60
## [85] 20.17 15.06 14.20 23.07 20.36 25.76 17.02 13.71 23.88 26.72 22.58 24.51
## [97] 45.23 38.07 36.97 35.17 37.83 43.69 29.14 24.56 25.20 19.27 20.88 18.27
## [109] 16.96 21.38 18.33 23.18 21.15 21.97 22.63 9.74 16.71 26.18 30.39 22.95
## [121] 25.51 20.28 16.86 21.94 20.59 21.64 27.42 35.72 23.47 31.57 23.71 19.01
## [133] 21.52 19.40 24.32 34.28 23.96 23.14 26.60 24.94 28.49 28.18 41.64 23.85
## [145] 28.67 28.76 35.16 35.46 28.74 26.99 31.74 40.41 33.73 25.57 29.13 29.74
## [157] 36.32 22.58 22.82 46.67 29.44 25.40 17.27 20.38 21.55 19.19 15.89 18.37
## [169] 30.51 18.47 11.70 18.45 24.75 16.63 20.80 19.62 22.56 19.87 20.22 21.16
## [181] 22.13 20.66 22.82 32.86 26.04 20.60 44.44 35.28 38.03 28.46 29.10 30.19
## [193] 26.17 22.71 23.39 23.44 16.27 21.26 24.67 19.12 23.26 21.75 24.59 27.26
## [205] 22.63 26.40 31.60 29.57 30.90 32.29 46.86 41.73 49.31 66.76
```

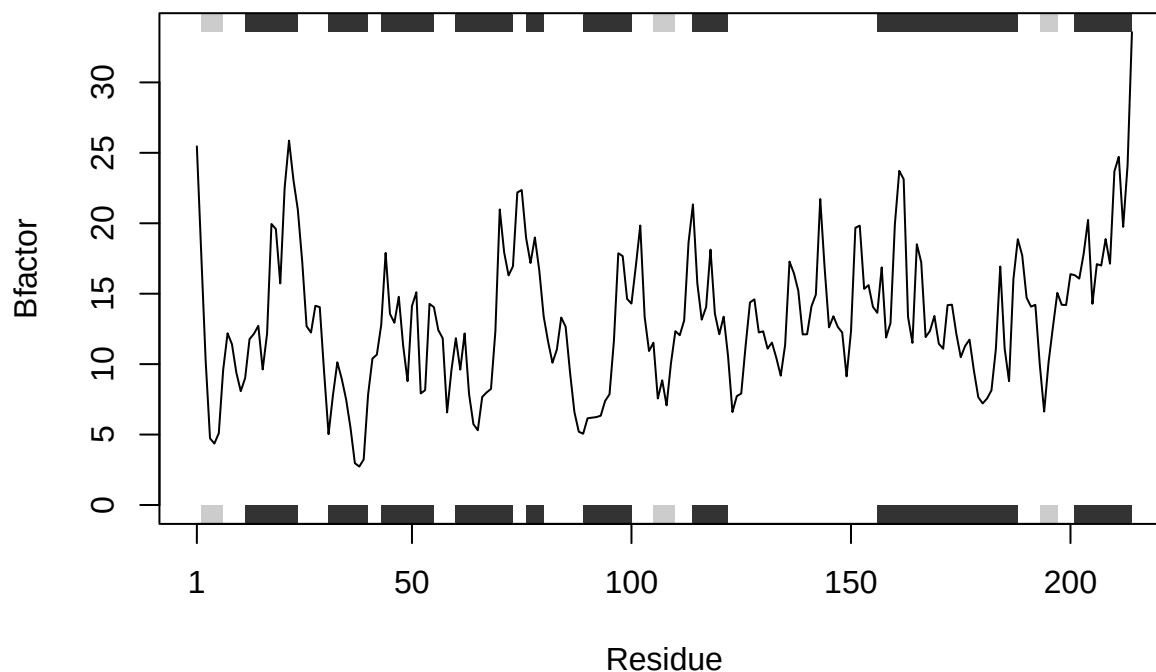
```
plot_protein("1E4Y")
```

```
## Note: Accessing on-line PDB file
```

```
## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
```

```
## /var/folders/1x/9jyvqlns09n6_zjnmq3xgx5h0000gn/T//RtmpzmuEmj/1E4Y.pdb exists.
```

```
## Skipping download
```



```
## [1] 25.46 17.86 10.28 4.73 4.36 5.10 9.59 12.19 11.41 9.39 8.08 9.01
## [13] 11.77 12.15 12.72 9.62 12.18 19.95 19.59 15.73 22.51 25.87 23.08 20.97
## [25] 17.28 12.69 12.24 14.14 14.05 9.38 5.03 7.78 10.13 8.96 7.50 5.48
```

```

## [37]  2.97  2.73  3.23  7.81 10.40 10.67 12.79 17.90 13.56 12.94 14.78 11.31
## [49]  8.79 14.13 15.10  7.92  8.15 14.28 14.04 12.42 11.84  6.57  9.59 11.84
## [61]  9.61 12.18  7.89  5.74  5.31  7.67  7.99  8.24 12.34 20.98 17.93 16.30
## [73] 16.94 22.19 22.36 18.96 17.18 18.99 16.65 13.39 11.61 10.10 11.03 13.31
## [85] 12.66  9.44  6.60  5.20  5.06  6.16  6.20  6.24  6.34  7.39  7.86 11.66
## [97] 17.87 17.67 14.63 14.30 16.98 19.84 13.36 10.93 11.52  7.56  8.85  7.07
## [109] 10.08 12.34 12.05 13.10 18.63 21.34 15.73 13.16 14.04 18.13 13.59 12.12
## [121] 13.37 10.57  6.60  7.73  7.91 11.31 14.38 14.60 12.25 12.33 11.10 11.53
## [133] 10.44  9.18 11.36 17.28 16.45 15.21 12.11 12.12 14.10 14.94 21.72 16.82
## [145] 12.61 13.40 12.64 12.24  9.13 12.31 19.68 19.83 15.34 15.61 14.07 13.64
## [157] 16.87 11.89 12.92 19.93 23.72 23.13 13.35 11.51 18.51 17.24 11.92 12.36
## [169] 13.42 11.45 11.09 14.19 14.22 12.15 10.49 11.29 11.74  9.53  7.65  7.21
## [181]  7.56  8.14 11.07 16.93 11.12  8.79 16.03 18.87 17.72 14.72 14.08 14.21
## [193]  9.99  6.63 10.11 12.64 15.06 14.21 14.20 16.39 16.31 16.07 17.83 20.24
## [205] 14.28 17.10 17.00 18.88 17.13 23.68 24.72 19.74 24.12 33.57

```

The `plot_protein()` function takes a PDB ID such as “4AKE” for the input and the optional arguments for the chain and atom type. The function serves to read a protein structure using the `read.pdb()` function which trims it to chain A alpha carbons using the trim function, `trim.pdb()`. This essentially extracts the B-factor values and plots to visualize the protein. Additionally, the function produces an output of a B-factor plot and returns a numeric vector of B-factor values.